

QING HOU

FILTERING NETWORK

Former Freight Yard Redevelopment
Southern New Orleans, Louisiana

Design Statement

The site lies in the lower Mississippi River Delta, where the river mouth has shifted over time. To sustain settlement and economic activity, the U.S. government and the U.S. Army Corps of Engineers constructed dams, canals, and other infrastructure near the former river outlet. These interventions altered flows and disrupted delta formation, reducing freshwater and sediment inputs. Saltwater intrusion has damaged low-salinity wetland vegetation, while industrial wastewater has degraded water quality and river ecosystems. Sea-level rise, wave action, and extreme weather have accelerated erosion, soil loss, and shoreline retreat.

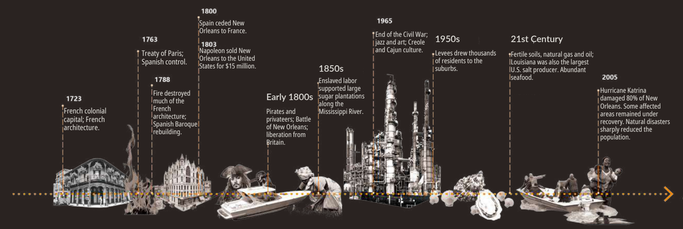
Without such intervention, sediment would accumulate as channels lengthen and flatten, eventually directing water into new channels. Abandoned channels would subside and evolve into lakes, wetlands, and shoals.

Respecting the site's cultural, economic, and environmental context, this proposal uses adaptive shorelines to integrate landscape production and protection while restoring delta conditions.

Filtering Network creates a connected ecological landscape that improves soil and water quality. Site-responsive regenerative design repurposes obsolete infrastructure and degraded areas, supporting long-term sustainability, continued adaptation, and the coexistence of living systems and natural processes within a self-renewing landscape. [9]

FILTER | Maintain the balance of soil and water.
PATH | Restore nature's tranquility.

Historical & Cultural Context



Location



United States > Louisiana > New Orleans / Mississippi Delta

Climate & Terrain Analysis



Slope & Aspect



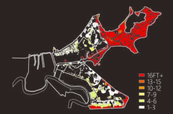
Levee Locations



Drainage & Outfalls



Land Change: 10 Years



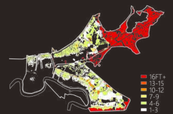
100-Year Flood



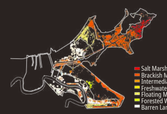
Coastal Vegetation: 10 Years



Land Change: 50 Years



500-Year Flood



Coastal Vegetation: 50 Years

Site & Surroundings



Mississippi Tributary



Small, Low-Quality Parks



Housing to the South



Debris & Mobile Structures



On-Site Rail Tracks



Poor Road Conditions



Nearby Parks



Elevation



Drainage & Outfalls



Master Plan

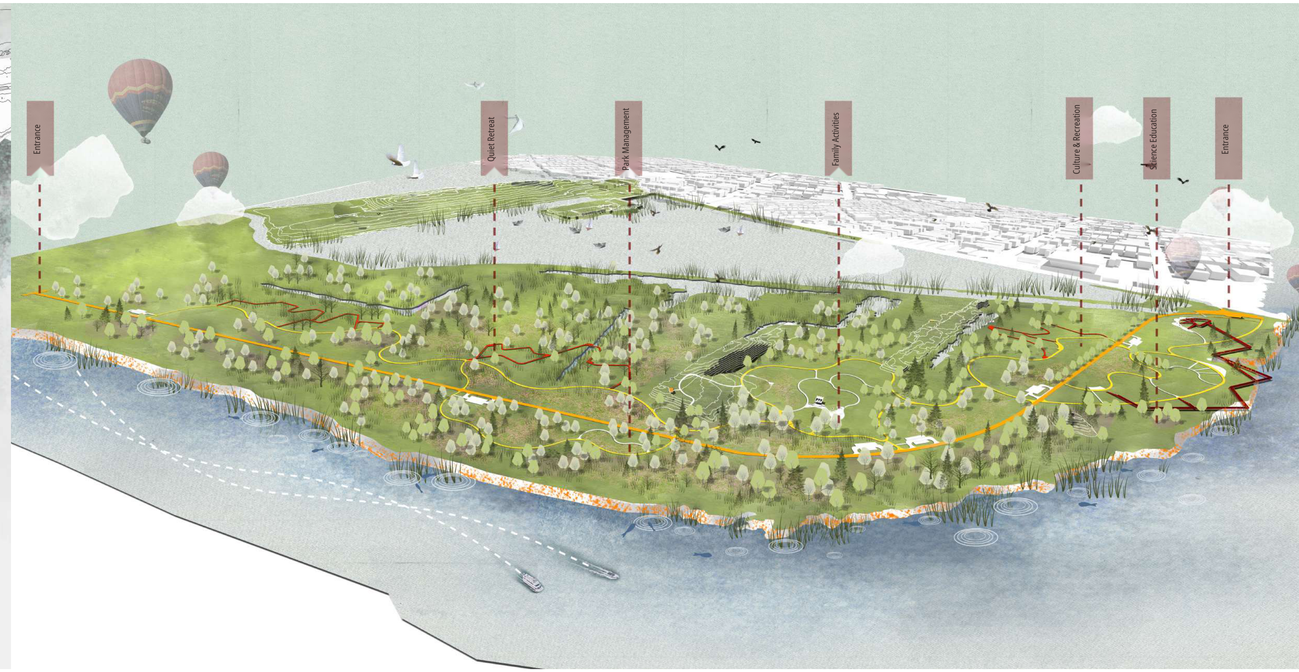


Main Entrance
Secondary Entrance
1. Entrance Plaza
2. Visitor Center
3. Elevated Promenade
4. Music Plaza
5. Plaza
6. Waterside Lookout
7. Parking
8. Shaded Flower Garden
9. Cultural Plaza
10. Elevated Landmark
11. Cultural Sculpture
12. Pergola
13. Viewing Plaza
14. Woodland Plaza
15. Woodland Fitness
16. Riverside Retreat
17. Rest Area
18. Meditation Corner
19. Overwater Lookout
20. Spring Garden
21. Landform
22. River Breeze Walk
23. Secluded Garden
24. Shaded Plaza
25. Children's Play Area
26. Rest Plaza
27. Woodland Water Garden
28. Family Camping
29. Green Slope
30. Supply Kiosk
31. Maple Grove
32. Spring Planting
33. Birdwatching Tower
34. Park Management
35. Boardwalk
36. Waterside Birdwatching
37. Flower Garden
38. Meadow
39. Woodland Trail
40. Hillside Walk
41. Viewing Point
42. Entrance Plaza

Table 3-1. Land Use Summary

Total Park Area (m^2)	Land Use			Area (m^2)	Share (%)
2,987,647	Land	Green Space	1,260,608 m^2 88.194%		
		Building Footprints	8,963 m^2 0.006%	1,432,846	100.000
		Paths & Paved Areas	172,955 m^2 11.800%		
		Water	1,554,801 m^2	1,554,801	100.000

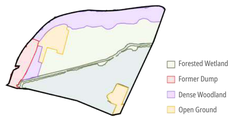
Aerial View



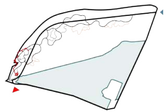
Site Layout Development



Existing Landform



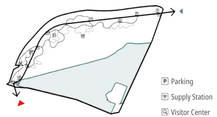
Existing Vegetation



Circulation



Functional Zones



Facilities

Activity Analysis



Camping



Cycling



Hiking

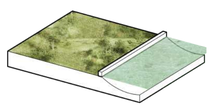


Walking

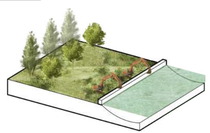


Birdwatching

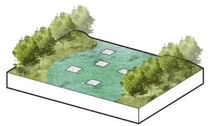
Riverbank Treatment



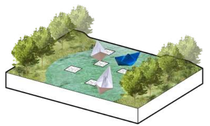
Existing Hard Levee



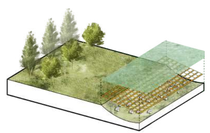
Remove Hard Levee



Oyster Farming



Collect Oyster Shells



Oyster-Reef Shoreline

Design Detail 1



Node 2: Detail Plan



Node 2: Location

1. Tree Pit
2. Landscape Boulder
3. Specimen Tree
4. Seating
5. Sculpture
6. Accessible Ramp
7. Camp Supply Station



B-B Section

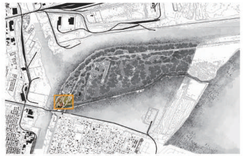




Design Detail 2



Node 3: Detail Plan



Node 3: Location

1. Elevated Pergola
2. Tree Pit
3. Science Display Wall
4. Hurricane Exhibit Prod
5. Music Wall
6. Sculpture
7. Walkway



C-C Section



